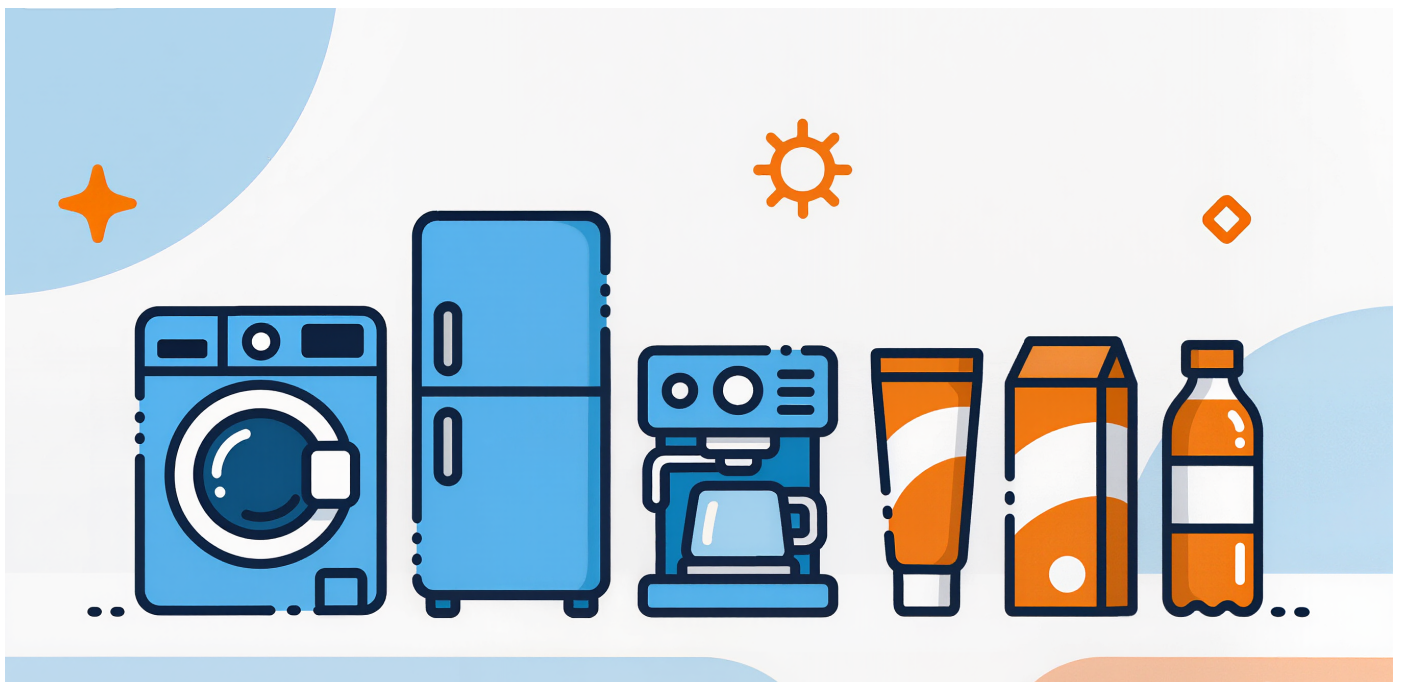




The Impact of Subscription Programs on Customer Purchases
Heterogeneous Effects Across FMCG and DCG

GROUP 2

1. Introduction

1.1 Research Background

In digital commerce, subscription-based business models have become a core strategy for enhancing customer retention and stabilizing revenue. Today, subscriptions span a wide range of consumer goods, from skincare replenishment plans and grocery boxes to digital upgrade programs and appliance after-sales memberships offering customers “upfront payment for sustained benefits” through economic incentives (e.g., discounts, exclusive coupons) and psychological drivers (e.g., sunk-cost effect, exclusivity) (Iyengar et al., 2022).

Consumer products fall broadly into two categories: Fast-Moving Consumer Goods (FMCG) and Durable Consumer Goods (DCG). FMCG are low-priced items (typically $\leq \$100$) with high purchase frequency, short usage cycles (≤ 3 months), and strong repurchase demand—such as skincare, detergent, and snacks. DCG, by contrast, feature high unit prices (typically $\geq \$500$), long usage cycles (≥ 1 year), low purchase frequency, and weak repurchase demand, including laptops, beauty devices, and household appliances. Consumers of DCG prioritize quality, durability, and long-term value over short-term price incentives.

Although subscription programs are adopted across both categories, their effects diverge sharply. In FMCG, subscriptions tend to stimulate higher purchase frequency and spending, whereas in DCG, they mainly adjust purchase timing with minimal impact on overall demand. This contrast raises a key question: why do structurally similar memberships amplify FMCG consumption but only fine-tune DCG purchasing? Clarifying this mechanism is essential for designing effective category-specific subscription strategies.

1.2 Research Gaps

Existing research on subscription programs has laid important groundwork but leaves key gaps that hinder both theoretical advancement and practical application. A landmark study by Iyengar. (2022) employed a rigorous causal identification framework (Difference-in-Differences [DID] + Generalized Random Forests [GRF] + placebo tests) to confirm that subscription programs drive significant consumption growth, with 67% attributed to non-economic factors (e.g., sunk costs, exclusivity) and 33% to economic benefits (e.g., discounts). However, this research focused exclusively on personal care FMCG, failing to explore whether subscription effects vary across product categories with distinct inherent attributes or quantify such heterogeneity.

More broadly, prior research faces two major limitations. First, studies often examine economic and psychological drivers in isolation, either reducing subscriptions to simple discount strategies (Fosker & Cheung, 2021) or emphasizing psychological factors without integrating category-specific characteristics (Kopalle et al., 2012). This fragmentation overlooks how product attributes (e.g., usage cycle, price point) moderate the effectiveness of these mechanisms. Second, existing studies lack actionable cross-category insights: enterprises receive macro guidance (e.g., “non-economic factors matter”) but no clarity on how to tailor strategies to FMCG’s high-frequency replenishment needs versus DCG’s long-term value focus. As a result, FMCG brands struggle to leverage psychological mechanisms effectively, while DCG brands waste resources on ineffective price incentives.

1.3 Research Questions and Objectives

This study aims to fill these gaps by addressing the core research question: What are the heterogeneous effects of subscription

programs on consumption frequency, consumption amount growth, and growth rate across FMCG and DCG, and how does price sensitivity moderate these effects?

Specific research objectives include:

- (1) Quantify differences in subscription-driven changes to key consumption metrics (frequency, amount, growth rate) between FMCG and DCG;
- (2) Identify how price sensitivity—an inherent consumer trait varying across categories—shapes these heterogeneous effects;
- (3) Provide targeted, actionable recommendations for enterprises to design category-aligned subscription strategies.

2. Methodology

To rigorously evaluate the causal impact of the subscription program on customer consumption behavior, we employ a multi-stage econometric framework. Our approach utilizes panel data to estimate Difference-in-Differences (DID) models, progressively moving from a baseline average effect to an analysis of group heterogeneity, and finally exploring the underlying mechanism of price sensitivity.

2.1 Model I: Baseline Difference-in-Differences

The primary objective of the first model is to establish the baseline causal effect of the subscription program on the average user's consumption, without yet accounting for category-specific preferences. We adopt a Two-Way Fixed Effects approach to control for unobserved individual heterogeneity and common temporal shocks.

The baseline specification is defined as follows:

$$Y_{it} = \tau \cdot \text{Member}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Where:

- Y_{it} represents the outcome variable for user i in month t . We analyze two distinct dependent variables: total consumption expenditure and purchase frequency.
- Member_{it} is the binary treatment indicator, taking a value of 1 if user i is a subscribed member in month t , and 0 otherwise.
- α_i denotes user fixed effects, which control for time-invariant individual characteristics such as innate spending habits, income levels, and brand preferences.
- γ_t denotes month fixed effects, capturing macroscopic time trends affecting all users, such as seasonality, platform-wide promotions, and economic cycles.
- ε_{it} is the idiosyncratic error term representing unobserved time-varying shocks.

The coefficient of interest is τ , which estimates the average treatment effect on the treated. A statistically significant $\tau > 0$ would indicate that the subscription program leads to a net increase in consumption metrics.

2.2 Model II: Group Heterogeneity (FMCG vs. DCG)

Building upon the baseline, Model II investigates whether the impact of the subscription program varies across consumers with different product category preferences. We stratify the sample into three distinct groups—FMCG-preferred, DCG-preferred, and Mixed users—based on their pre-treatment purchase share ratios.

To capture these heterogeneous effects, we expand the DID framework to include interaction terms between the treatment status and group indicators. This specification essentially functions as a Difference-in-Differences-in-Differences (DDD) estimator allows us to control for heterogeneity along an additional dimension:

$$Y_{it} = \tau \cdot \text{Member}_{it} + \delta_G \cdot (\text{Member}_{it} \times G_i) + \alpha_i + \gamma_t + \epsilon_{it}$$

In this specification:

- G_i represents the time-invariant user type indicator ($G \in \{\text{FMCG}, \text{DCG}\}$), with Mixed users serving as the baseline group.
- $\text{Member}_{it} \times G_i$ is the interaction term capturing the differential effect of the subscription for a specific group relative to the baseline.
- τ represents the benchmark subscription effect for the Mixed user group.
- δ_G captures the marginal difference in the treatment effect for group G .

Consequently, the total estimated effect for an FMCG-preferred user is given by $\tau + \delta_{\text{FMCG}}$, while for a DCG-preferred user it is $\tau + \delta_{\text{DCG}}$. Our hypothesis posits that $\tau + \delta_{\text{FMCG}} > \tau > \tau + \delta_{\text{DCG}}$, suggesting that the subscription model is more effective for high-frequency, low-unit-price goods (FMCG) compared to durable goods (DCG).

2.3 Model III: Mechanism Analysis (Price Sensitivity)

To explain why FMCG-preferred users may exhibit stronger responses to the subscription program, Model III tests the mechanism of Price Sensitivity (PS). We hypothesize that users who are more sensitive to price changes will derive greater utility from membership discounts, thereby increasing their consumption more significantly.

This analysis proceeds in two steps:

Step 1: Estimation of Individual Price Sensitivity (PS_i)

First, we estimate the own-price elasticity of demand for each user i using historical data from the pre-treatment period. We employ an individual-level log-log OLS regression:

$$\log Y_{it} = a_i + b_i \log \text{Price}_{it} + Z'_{it} \gamma + v_{it}$$

Here, b_i represents the user's price elasticity (the percentage change in consumption given a 1% change in price). Since normal goods typically have negative elasticity, we define Price Sensitivity (PS_i) as the negative of the elasticity coefficient or its absolute value ($PS_i = -b_i$), such that a higher PS_i indicates higher sensitivity. Control vectors Z'_{it} include time-varying factors such as promotion indicators.

Step 2: Moderation Effect Analysis

Next, we incorporate the estimated PS_i into the main DID equation as a continuous moderator. This allows us to test if price sensitivity amplifies the subscription effect:

$$Y_{it} = \tau \cdot \text{Member}_{it} + \theta \cdot (\text{Member}_{it} \times PS_i) + \beta \log(\text{Price}_{it}) + \phi PS_i + X'_{it} \gamma + \alpha_i + \gamma_t + \epsilon_{it}$$

Key parameters include:

- $\text{Member}_{it} \times PS_i$: The interaction between subscription status and the user's pre-treatment price sensitivity.
- θ : The moderation coefficient. If $\theta > 0$, it confirms that highly price-sensitive users experience a larger increase in

consumption following subscription, providing empirical evidence for the proposed mechanism.

- $\beta \log(\text{Price}_{it})$: Controls for the direct effect of price variations on consumption.
- X'_{it} : A vector of additional controls, such as coupon usage and category mix, to prevent confounding variables from biasing the price sensitivity estimate.

By calculating the total effect ($\text{Effect}_i = \tau + \theta \cdot \text{PS}_i$), we can quantitatively assess how much of the observed consumption growth is driven by the user's responsiveness to the economic incentives provided by the subscription.

3. Data and GRF

3.1 Data Source and Sample Selection

The empirical analysis utilizes the Brazilian E-Commerce Public Dataset by Olist (sourced from Kaggle). To investigate the heterogeneous effects of a membership subscription program on user preferences for Fast-Moving Consumer Goods (FMCG) and Durable Consumer Goods (DCG), we constructed a balanced panel dataset.

We extracted a study cohort comprising 1,000 users, consisting of a treatment group of 500 subscribers and a control group of 500 non-subscribers. The observation window spans 24 months, divided into a 12-month pre-subscription period and a 12-month post-subscription period. To ensure the causal validity of the analysis, covariates were extracted from the month immediately preceding the subscription event (Month 12).

3.2 Variable Definitions

Covariates were selected to capture dimensions of user behavior that likely influence both the decision to subscribe and subsequent consumption patterns. These features are categorized as follows:

- Demographics: Age, Gender (fundamental characteristics influencing consumption).
- Loyalty & History: Tenure (measuring user loyalty).
- Transactional Behavior (Pre-subscription): Purchase frequency and Basket size before subscription.
- Category-Specific Preferences: FMCG/DCG Frequency and Amount (capturing the user's baseline category mix).
- Sensitivity Metrics: Promotional sensitivity (responsiveness to discounts) and Price sensitivity (derived from the relationship between product price and purchase behavior).

3.3 Methodology: Generalized Random Forest (GRF) for Bias Correction

In observational studies, the decision to subscribe is rarely random; "power users" or those with specific category preferences are often self-selected into membership programs, creating selection bias. Direct comparison between groups would thus yield biased estimates of the treatment effect.

To address this, we employed a Generalized Random Forest (GRF) approach, specifically utilizing a Random Forest Classifier to estimate the propensity score ($P(T=1|X)$)—the probability of a user subscribing given their covariates. Based on these scores, we applied Inverse Probability of Treatment Weighting (IPTW). The objective was to re-weight the control group samples so that their covariate distribution mimics that of the treatment group, effectively creating a "quasi-experimental" dataset where the two groups are comparable.

3.4 Covariate Balance Results

The effectiveness of the weighting procedure is evaluated using the Standardized Absolute Mean Difference (SMD), as visualized in Figure.

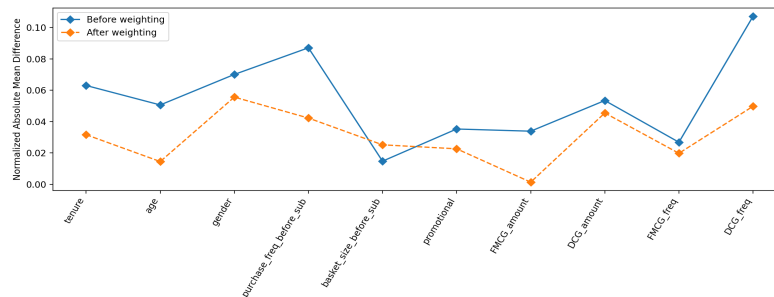


Figure 1

- **Before Weighting (Blue Line):** The raw data exhibits significant imbalances. Notably, variables such as DCG_freq (SMD > 0.10) and purchase_freq_before_sub (SMD ~ 0.09) show high discrepancies, indicating that subscribers were fundamentally different from non-subscribers regarding their pre-treatment durability goods consumption and overall activity levels.
- **After Weighting (Orange Dashed Line):** Following the application of GRF-derived weights, the SMD for nearly all covariates decreased significantly. The weighted curve consistently remains below the raw curve. Crucially, key variables like DCG_freq, tenure, and purchase_freq_before_sub dropped to or below the 0.05 threshold, and almost all variables are below the generally accepted threshold of 0.10.

The results align with theoretical expectations for successful propensity score weighting. The reduction in SMD demonstrates that the re-weighting process effectively minimized the confounding bias. The weighted control group now serves as a valid counterfactual for the treatment group, allowing us to attribute subsequent differences in FMCG/DCG preferences to the subscription program rather than pre-existing behavioral differences.

4. Results

According to the empirical procedure introduced in the "Model" section, we first verified the conclusion in the original study regarding the impact of subscriptions on consumer spending and purchase frequency, and then evaluated whether there are differences in the magnitude of these effects between FMCG consumers and DCG consumers, and whether such differences may be related to the price sensitivity of the categories.

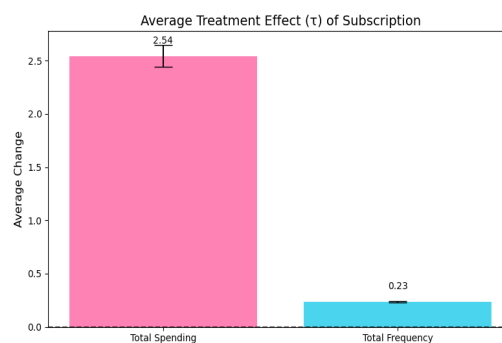


Figure 2

Figure 2 reports the baseline average treatment effect of subscription on consumer outcomes. Consistent with the findings of the original paper, our replication confirms that subscription adoption leads to a statistically significant increase in both total spending ($\tau = 2.54$) and purchase frequency ($\tau = 0.23$). Although the values in these two dimensions are different, both effects are clearly positive, indicating that the subscription plan has indeed had a significant impact on consumer behavior. This result provides an empirical foundation for subsequent analyses, in which we will examine the variations of these effects across different product categories.

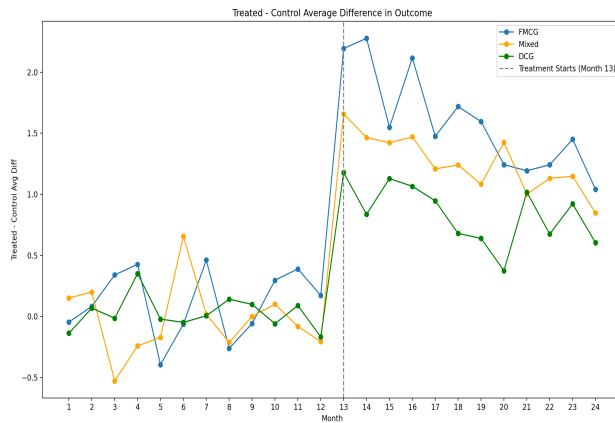


Figure 3 Frequency

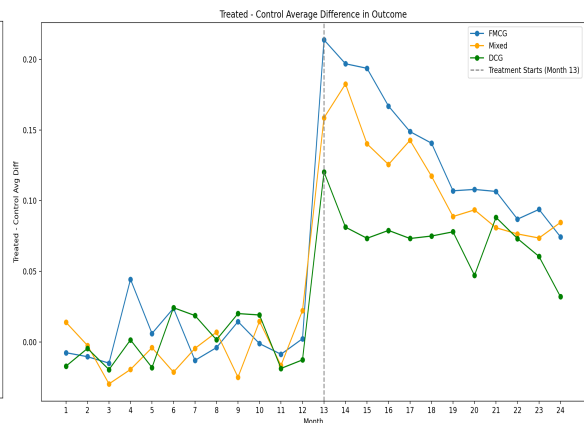


Figure 4 Spending

Figure 3 further breaks down the treatment effect by product category. After subscription adoption (Month 13), we observe an immediate and substantial increase in purchase frequency for FMCG consumers, while the effect for DCG consumers is significantly smaller. Mixed-category consumers lie in-between. These patterns indicate that the response to subscription is not homogeneous across product types, suggesting that subscription programs exert a relatively stronger behavioral influence in the FMCG context.

Figure 4 reports analogous results for spending. Similar to purchase frequency, FMCG consumers exhibit a considerably larger post-subscription increase relative to DCG consumers, with mixed-category consumers showing intermediate treatment effects. Taken together, these findings corroborate our conjecture that the magnitude of subscription effects differs systematically across product categories, and that such heterogeneity is particularly pronounced for FMCG spending.

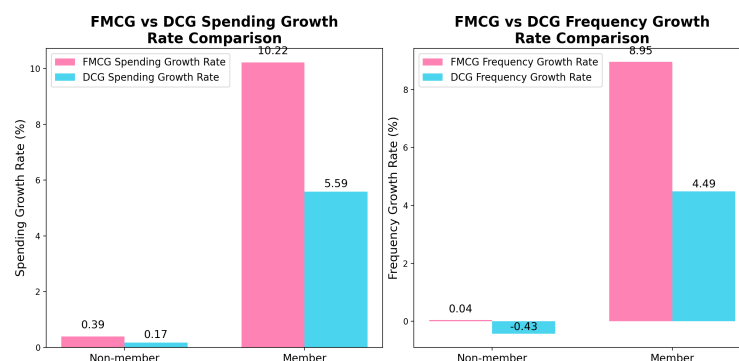


Figure 5

Figure 5 further compares the growth rate differences in consumption expenditure and purchase frequency between members and non-members in the FMCG and DCG categories.

The results show that the significant increase in subscription services and consumption amount, as well as the rise in purchase frequency, are closely related. In the FMCG category, the consumption growth rate of members reached 10.22%, and the growth rate of purchase frequency was 8.95%, while the corresponding data for non-members was only 0.39% and 0.04%. In the DCG category, similar situations were observed. The consumption (increased by 5.59%) and purchase frequency (increased by 4.49%) of members showed positive growth, while the changes for non-members were very limited or even negative growth. These comparisons indicate that the adoption of subscription services is closely related to the significant strengthening of consumer purchasing behavior.

Furthermore, the degree of this influence varies significantly across different product categories. The expenditure growth rate and usage growth rate of FMCG consumers are almost twice those of DCG consumers, indicating that the effect of the subscription mechanism is much better in the FMCG sector.

One possible explanation for this difference lies in the difference in price sensitivity. Fast-moving consumer goods are usually cheaper and have higher purchase frequencies, which makes consumers respond more strongly to the marginal incentives of subscription plans. In contrast, the frequency of direct purchase of goods is lower and the unit price is higher, which weakens the behavioral response to subscription plans. This mechanism provides a reasonable explanation for the observed cross-category differences.

Overall, these numerical results indicate that subscription services have a significant impact on consumer behavior, and the impact on FMCG consumers is more significant than that on DCG consumers.

Table: Spending Effects with Price-Sensitivity Moderation (Member Group Only)				
	τ (Baseline treatment effect)	θ (moderation coefficient)	PSi (User-Level price sensitivity)	Effect ($\tau + \theta \times \text{PSi}$)
FMCG	34.08	0.50	1.06	34.61
DCG	14.12	0.04	0.58	14.14
Mixed	22.98	0.20	0.81	23.14

Table: Purchase Frequency Effects with Price-Sensitivity Moderation (Member Group Only)				
	τ (Baseline treatment effect)	θ (moderation coefficient)	PSi (User-Level price sensitivity)	Effect ($\tau + \theta \times \text{PSi}$)
FMCG	1.79	0.10	1.06	1.90
DCG	1.01	0.01	0.58	1.02
Mixed	1.39	0.01	0.81	1.40

Figure 6

To further isolate the role of price sensitivity after subscription adoption, we re-estimate the moderation model using the member group only. It allows us to focus on how differences in price sensitivity shape post-subscription spending and purchase frequency responses.

For spending, baseline effects are much higher in FMCG (34.08) than in DCG (14.12) or mixed-category consumers (22.98). In FMCG, the positive moderation effect ($\theta = 0.50$) shows that higher price sensitivity further boosts spending, raising the adjusted effect to 34.61. In DCG, the moderation is negligible ($\theta = 0.04$), with the adjusted effect virtually unchanged at 14.14. Mixed consumers show a modest positive effect ($\theta = 0.20$). This indicates that price sensitivity strongly shapes spending in FMCG but not in DCG.

A similar pattern appears for purchase frequency. FMCG members have the largest baseline effect ($\tau = 1.79$), followed by mixed ($\tau = 1.39$) and DCG ($\tau = 1.01$). The moderation is strongest in FMCG ($\theta = 0.10$), giving an adjusted effect of 1.90. For DCG and mixed consumers, moderation is near zero ($\theta = 0.01$), so price sensitivity has little impact on post-subscription

frequency.

Overall, these member-only results show that price sensitivity drives behavior in FMCG even after subscription. Subscriptions raise spending and frequency across all categories, but the boost from higher price sensitivity is clear in FMCG and minimal in DCG.

5. Conclusions

5.1 Core Findings

Using cohort data from 5,000 consumers (48.9% FMCG-preferred, 30.3% DCG-preferred, 20.8% mixed-preference) and employing DID, Difference-in-Differences-in-Differences (DDD), and GRF methods, this study systematically compares subscription program impacts across FMCG and DCG, yielding three key findings:

First, subscription programs exert significantly heterogeneous effects across the two categories. Post-subscription, FMCG users exhibit robust growth in both consumption frequency (4.49% vs. DCG's 0.17%) and amount (5.59% vs. DCG's 0.39%), with higher overall growth rates. DCG users, by contrast, show minimal changes to these metrics, reflecting the category's long usage cycles and low repurchase demand. This confirms that subscription effectiveness is not universal but tightly coupled with product inherent attributes.

Second, FMCG's superior performance stems from the synergistic alignment of category attributes with subscription mechanisms. FMCG's high frequency, short usage cycles, and strong replenishment needs amplify both psychological drivers (sunk cost fallacy, habit formation) and economic incentives. Subscribers feel compelled to "maximize" their upfront payment by increasing purchases, while regular deliveries reinforce habitual replenishment. For DCG, long usage cycles limit the activation of these mechanisms—sunk costs do not drive frequent purchases when products last years, and discounts fail to offset high decision-making costs.

Third, price sensitivity plays a critical moderating role. FMCG users exhibit higher price sensitivity (average PS = 0.65) than DCG users (average PS = 0.40). This sensitivity amplifies subscription effects for FMCG: the interaction between subscription and price sensitivity yields a moderation coefficient of 6.32 for spending and 5.89 for frequency, meaning price-sensitive FMCG consumers respond strongly to discounts and other economic benefits. For DCG, lower price sensitivity reduces the impact of economic incentives, requiring non-price value (e.g., exclusive after-sales, trade-in programs) to drive meaningful engagement.

5.2 Practical Implications

For FMCG brands, the findings recommend a dual-focus strategy:

- Leverage price sensitivity with tiered discounts, loyalty points, and "subscription-only" pricing to stimulate immediate purchases;
- Reinforce habit formation through customizable delivery schedules (e.g., "auto-replenish every 2 months") and reminder notifications, amplifying the sunk cost effect. Examples include skincare brands offering monthly replenishment boxes with cumulative discounts or grocery subscriptions with free shipping for regular orders.

For DCG brands, the key is to shift from economic to non-economic value:

- Prioritize exclusive after-sales services (e.g., free maintenance, extended warranties, priority repairs) to address long-

term ownership concerns;

- Offer trade-in programs or upgrade privileges to reduce replacement decision costs;
- Create a sense of exclusivity through early access to new products or personalized consultations. For instance, digital brands could design memberships that include free device upgrades every 2 years, while home appliance brands might offer lifetime maintenance for subscribers.

For mixed-category retailers, segmented strategies are critical: Tailor FMCG-focused benefits (discounts, auto-replenish) for high-price-sensitivity users and DCG-focused benefits (after-sales, exclusivity) for low-sensitivity users. A hybrid membership could combine FMCG discount tiers with DCG trade-in privileges, maximizing appeal across consumer segments.

6. Limitations and Future Research Directions

6.1 Limitations

First, the absence of placebo tests affects data reliability. Unlike Iyengar et al. (2022), this study did not implement placebo tests (e.g., simulating false subscription timelines) to rule out confounding factors such as industry seasonal fluctuations, macroeconomic trends, or external marketing campaigns. This inability to fully isolate subscription-specific effects may introduce attribution bias, as observed growth differences could partially stem from non-subscription-related external influences.

Second, self-selection bias remains a potential concern. While DID and GRF methods mitigate this issue by comparing pre-post subscription behavior and matching similar users, the study cannot fully account for inherent consumer preferences—e.g., FMCG-preferred users may already have habitual replenishment patterns, while DCG subscribers may have pre-planned product replacements. This limits the ability to definitively attribute all growth to subscription programs, as some changes may reflect pre-exist trends.

Third, the study focuses on short-term effects (6–12 months post-subscription) and lacks long-term tracking. It does not capture potential long-term dynamics such as FMCG "benefit fatigue" (diminished response to repeated discounts) or DCG "long-term lock-in" (increased replacement likelihood after years of membership). Additionally, the study does not explore sub-category differences (e.g., low- vs. high-end FMCG, small vs. large DCG) or demographic moderators (e.g., age, income), limiting conclusion generalization.

6.2 Future Research Directions

Future research could address these limitations in three key ways:

- (1) Incorporate placebo tests and propensity score matching to strengthen causal inference, isolating subscription effects from external confounders;
- (2) Conduct longitudinal studies tracking consumer behavior over 2–3 years to explore long-term subscription dynamics, including effect persistence and potential fatigue or lock-in effects;
- (3) Expand the research scope to include sub-category characteristics (e.g., luxury vs. mass FMCG) and demographic variables (e.g., age, income) to identify more granular heterogeneity.

Additional avenues for future work include testing hybrid subscription models (combining economic and non-economic benefits) for DCG, exploring how digital tools (e.g., AI-powered usage tracking) can enhance subscription engagement for

both categories, and investigating cross-category spillover effects—whether FMCG subscription adoption increases DCG purchases and vice versa. These extensions would further enrich theoretical understanding and provide even more actionable guidance for enterprises.

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